

Homework #4 (Submit before 09-May)

❖ Ch2_No.17

If the radius of the beam waist of the Gaussian beam is $w_0=0.3$ mm and the wavelength is $\lambda=632.8$ nm. Calculate the q parameter at the waist, 30 cm away from the beam waist, and the infinite distance from the beam waist, respectively.

❖ Ch2_No.22

The wavelength of the CO₂ laser is $\lambda=10.6$ μm , and the radius of the beam waist is $w_0=3$ mm. If we use a convex lens with $F=2$ cm to focus the laser beam, then where the position of lens should be if the radius of the focused beam is $w_0'=20$ μm and 2.5 μm .

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❖ Ch2_No.23

As shown in the figure below, the wavelength of the incident light beam is $\lambda = 10.6 \mu\text{m}$. Calculate the value of w_0'' and l_3 .

